Lecture 05

Nuclear Magnetic Resonance Spectroscopy

Introduction to NMR Spectroscopy

- Nuclear magnetic resonance spectroscopy is a powerful analytical technique used to characterize organic molecules by identifying carbon-hydrogen frameworks within molecules.

- The technique is basically another form of absorption spectroscopy, very similar in principle to infrared or ultraviolet spectroscopy.

- Two common types of NMR spectroscopy are used to characterize organic structure: ^{1}H NMR is used to determine the type and number of H atoms in a molecule; ^{13}C NMR is used to determine the type of carbon atoms in the molecule.

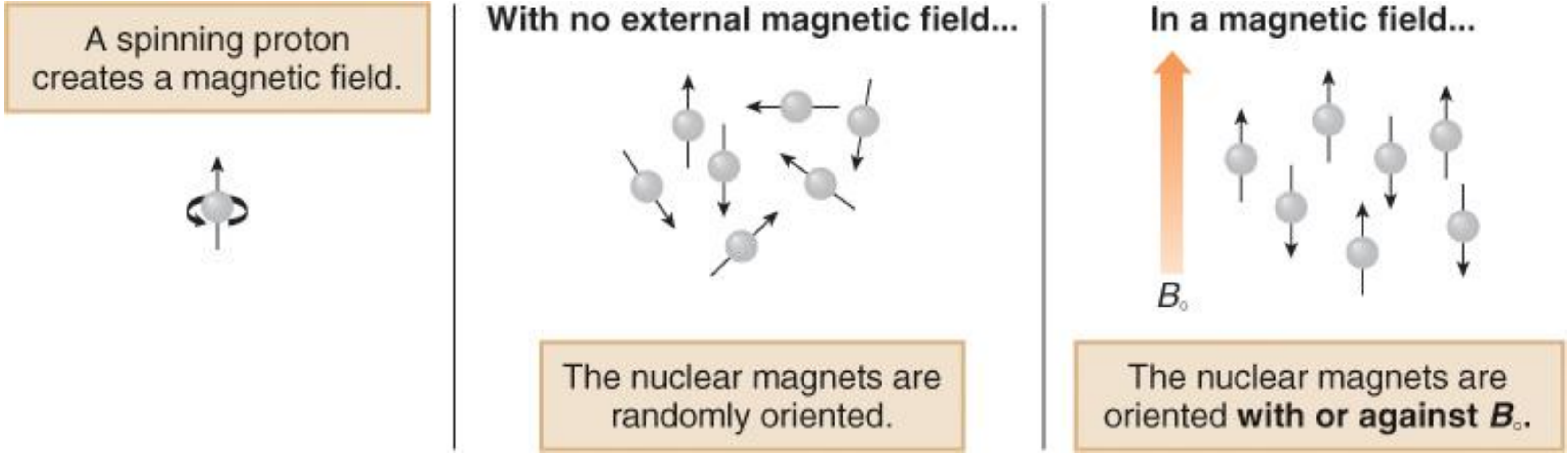
- The source of energy in NMR is radio waves which have long wavelengths, and thus low energy and frequency.

- When low-energy radio waves interact with a molecule, they can change the nuclear spins of some elements, including ^{1}H and ^{13}C.

- When a charged particle such as a proton spins on its axis, it creates a magnetic field. Thus, the nucleus can be considered to be a tiny bar magnet.

- Normally, these tiny bar magnets are randomly oriented in space. However, in the presence of a magnetic field B_0, they are oriented with or against this applied field. More nuclei are oriented with the applied field because this arrangement is lower in energy.

- The energy difference between these two states is very small (<0.1 cal).

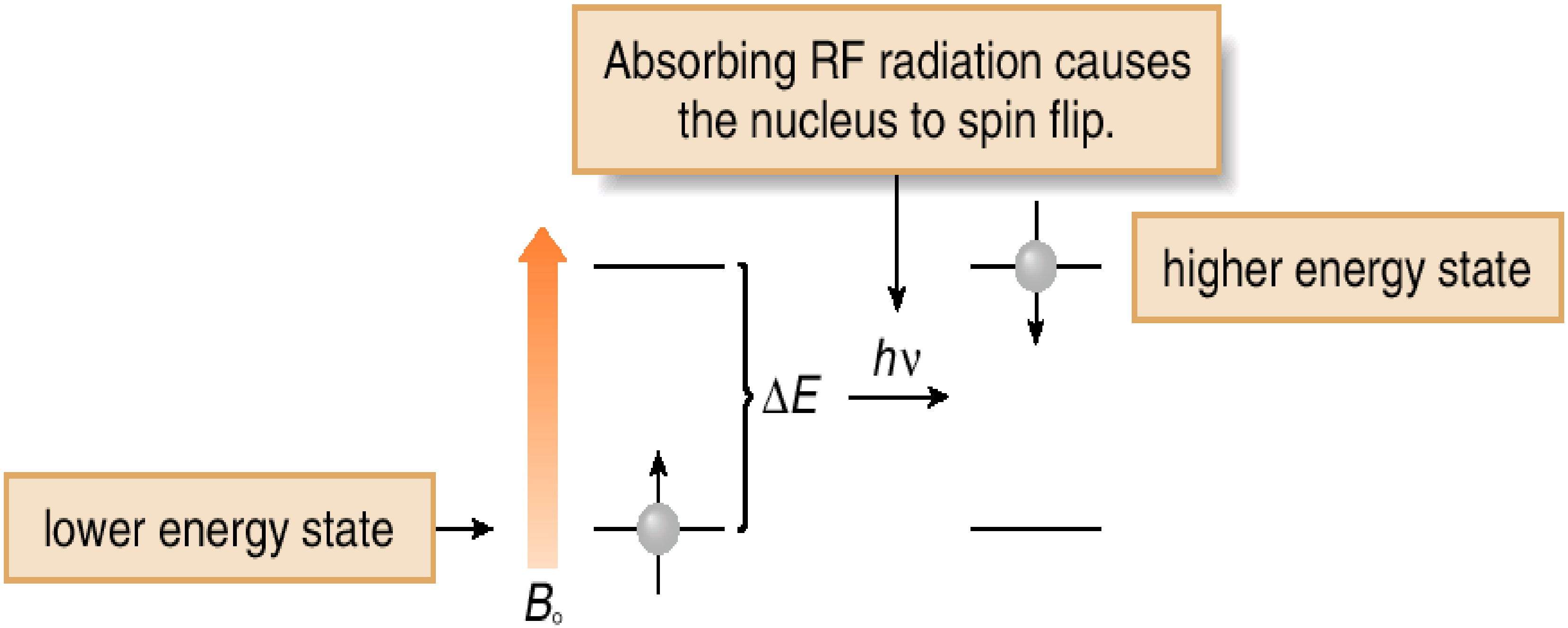


- In a magnetic field, there are now two energy states for a proton: a lower energy state with the nucleus aligned in the same direction as B_0, and a higher energy state in which the nucleus aligned against B_0.

- When an external energy source ($h\nu$) that matches the energy difference (ΔE) between these two states is applied, energy is absorbed, causing the nucleus to “spin flip” from one orientation to another.

- The energy difference between these two nuclear spin states corresponds to the low frequency RF region of the electromagnetic spectrum.

- Thus, two variables characterize NMR: an applied magnetic field B_0, the strength of which is measured in tesla (T), and the frequency ν of radiation used for resonance, measured in hertz (Hz), or megahertz (MHz)—(1 MHz = 10^6 Hz).

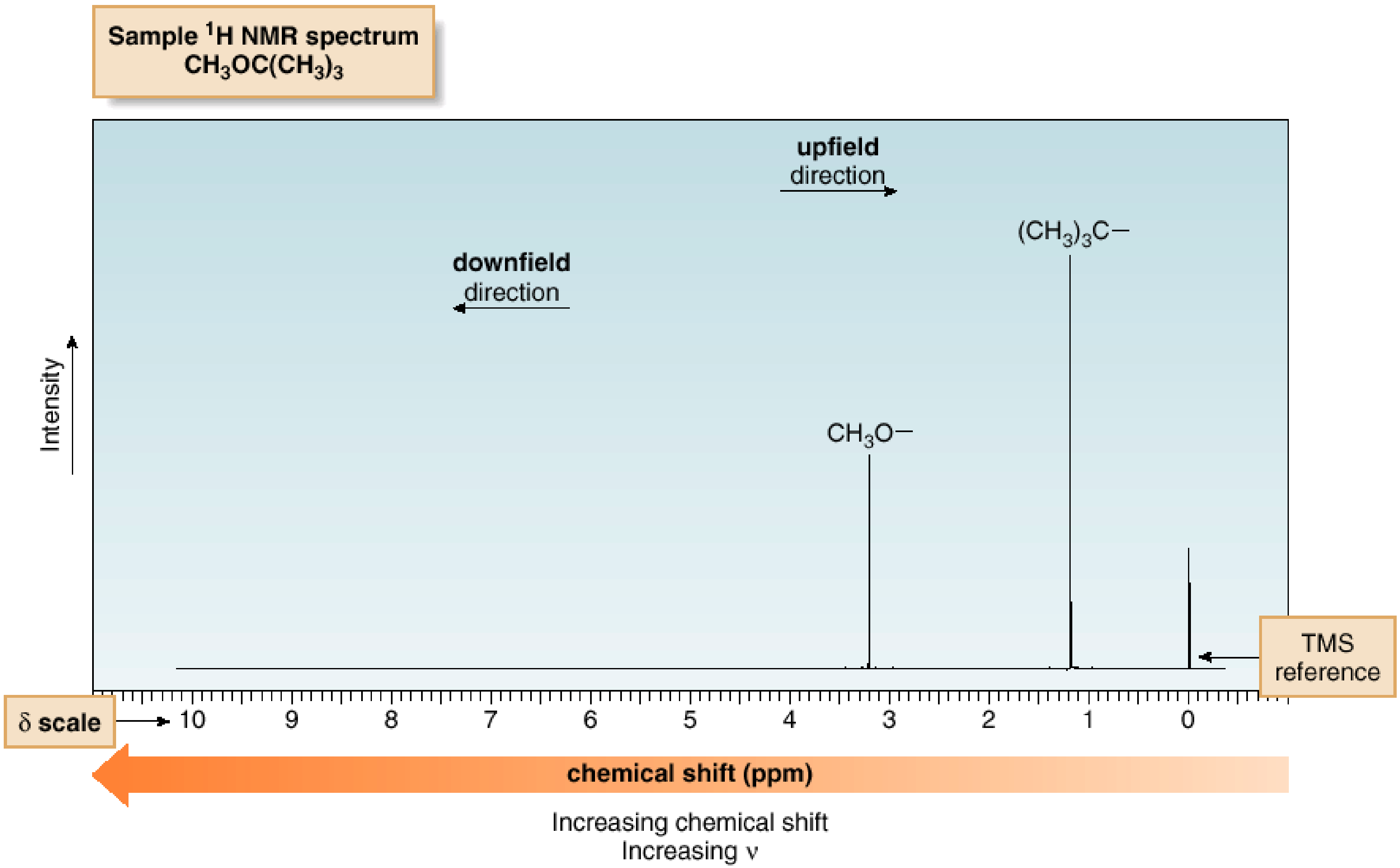


- A nucleus is in *resonance* when it absorbs RF radiation and "spin flips" to a higher energy state.

- Protons in different environments absorb at slightly different frequencies, so they are distinguishable by NMR.
- The frequency at which a particular proton absorbs is determined by its electronic environment.
- The size of the magnetic field generated by the electrons around a proton determines where it absorbs.
- Modern NMR spectrometers use a constant magnetic field strength B_0, and then a narrow range of frequencies is applied to achieve the resonance of all protons.
- Only nuclei that contain odd mass numbers (such as ^{1}H, ^{13}C, ^{19}F and ^{31}P) or odd atomic numbers (such as ^{2}H and ^{14}N) give rise to NMR signals.

^{1}H NMR—The Spectrum

- **An NMR spectrum is a plot of the intensity of a peak against its chemical shift, measured in parts per million (ppm).**



^{1}H NMR—The Spectrum

- NMR absorptions generally appear as sharp peaks.

- Increasing chemical shift is plotted from left to right.

- Most protons absorb between 0-10 ppm.

- The terms “upfield” and “downfield” describe the relative location of peaks. Upfield means to the right. Downfield means to the left.

- NMR absorptions are measured relative to the position of a reference peak at 0 ppm on the δ scale due to tetramethylsilane (TMS). TMS is a volatile inert compound that gives a single peak upfield from typical NMR absorptions.

^{1}H NMR—The Spectrum

- The chemical shift of the *x* axis gives the position of an NMR signal, measured in ppm, according to the following equation:

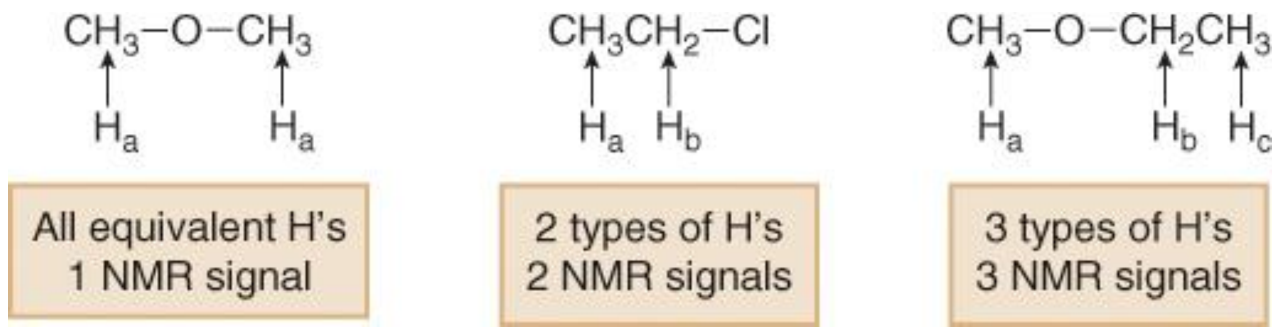
$$\underset{\text{(in ppm on the } \delta \text{ scale)}}{\textbf{chemical shift}} = \frac{\text{observed chemical shift (in Hz) downfield from TMS}}{\nu \text{ of the NMR spectrometer (in MHz)}}$$

- By reporting the NMR absorption as a fraction of the NMR operating frequency, we get units, ppm, that are independent of the spectrometer.
- Four different features of a ^{1}H NMR spectrum provide information about a compound's structure:

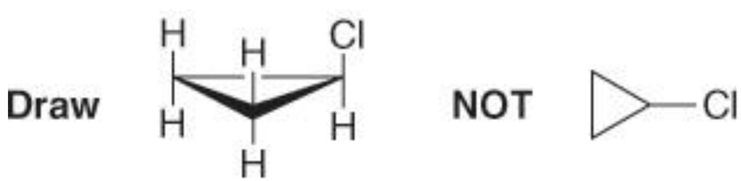
   A. Number of signals
   B. Position of signals
   C. Intensity of signals.
   D. Spin-spin splitting of signals.

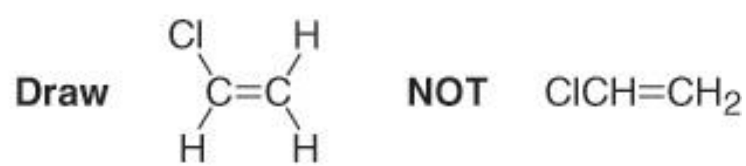
A. ^{1}H NMR—Number of Signals

- The number of NMR signals equals the number of different types of protons in a compound.
- Protons in different environments give different NMR signals.
- Equivalent protons give the same NMR signal.



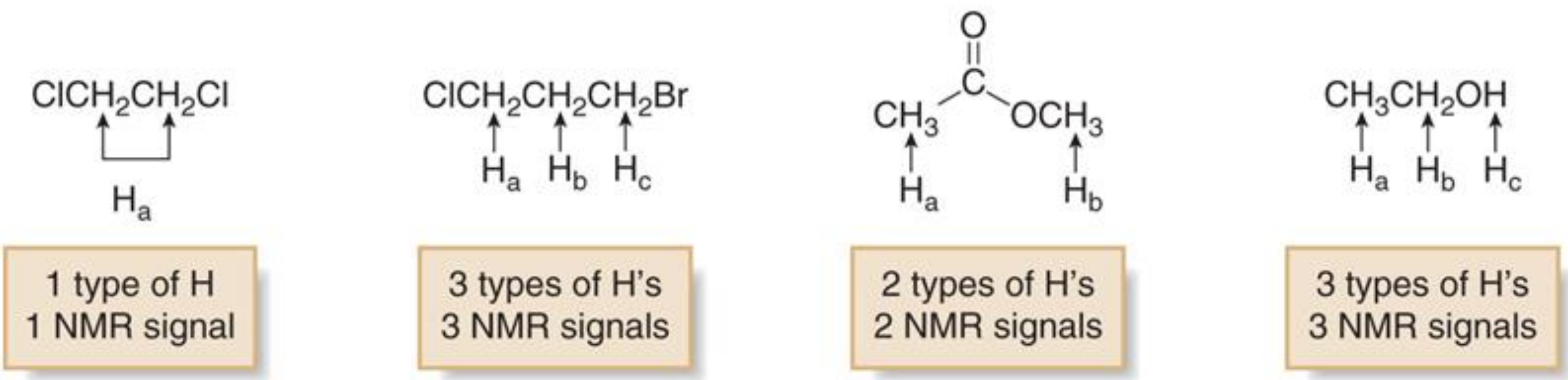
- To determine equivalent protons in cycloalkanes and alkenes, always draw all bonds to hydrogen.





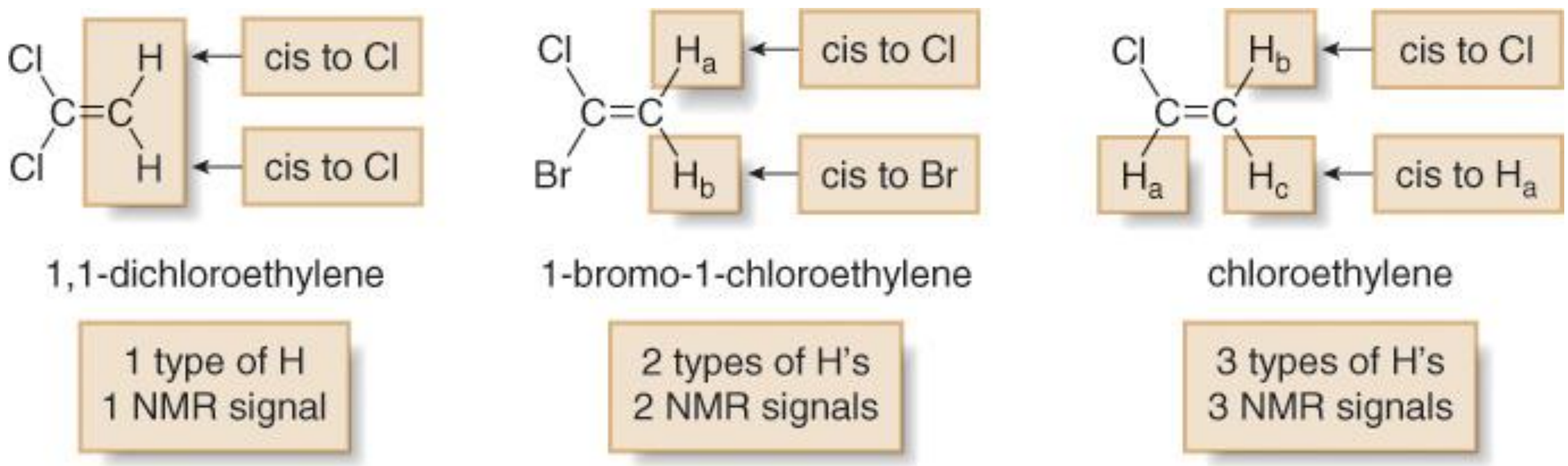
^{1}H NMR—Number of Signals

The number of ^{1}H NMR signals of some representative organic compounds



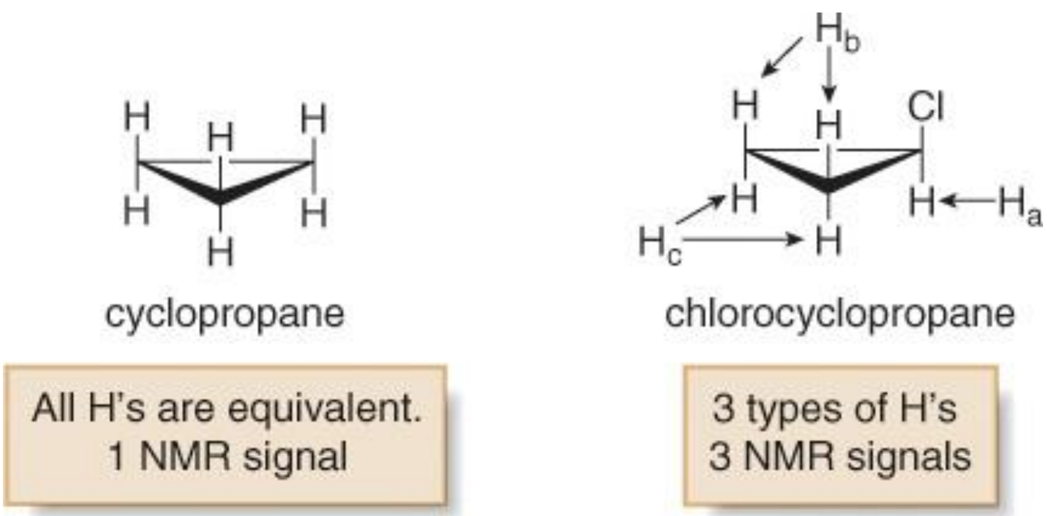
^{1}H NMR—Number of Signals

- **In comparing two H atoms on a ring or double bond, two protons are equivalent only if they are cis (or trans) to the same groups.**

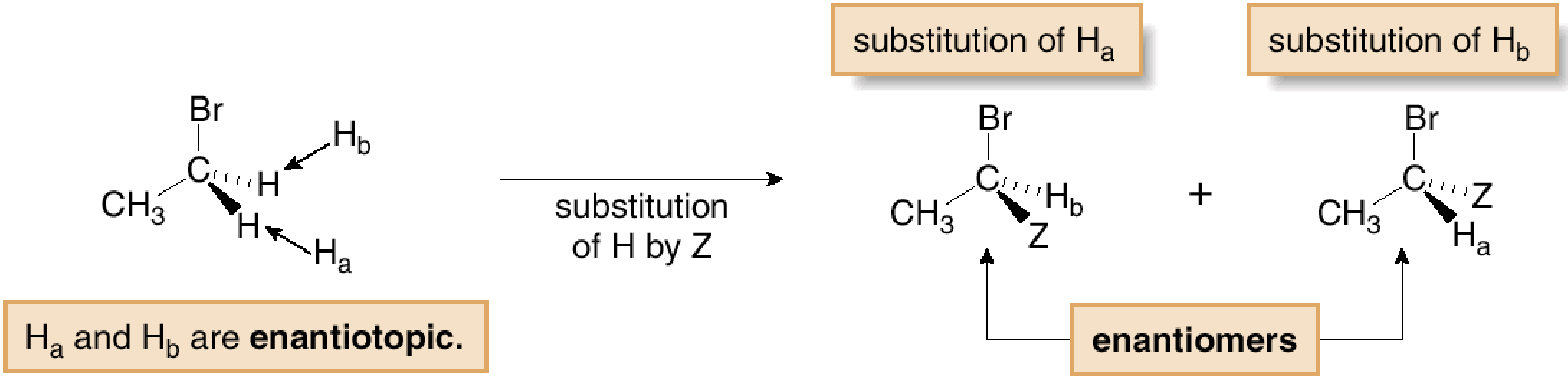


^{1}H NMR—Number of Signals

- **Proton equivalency in cycloalkanes can be determined similarly.**



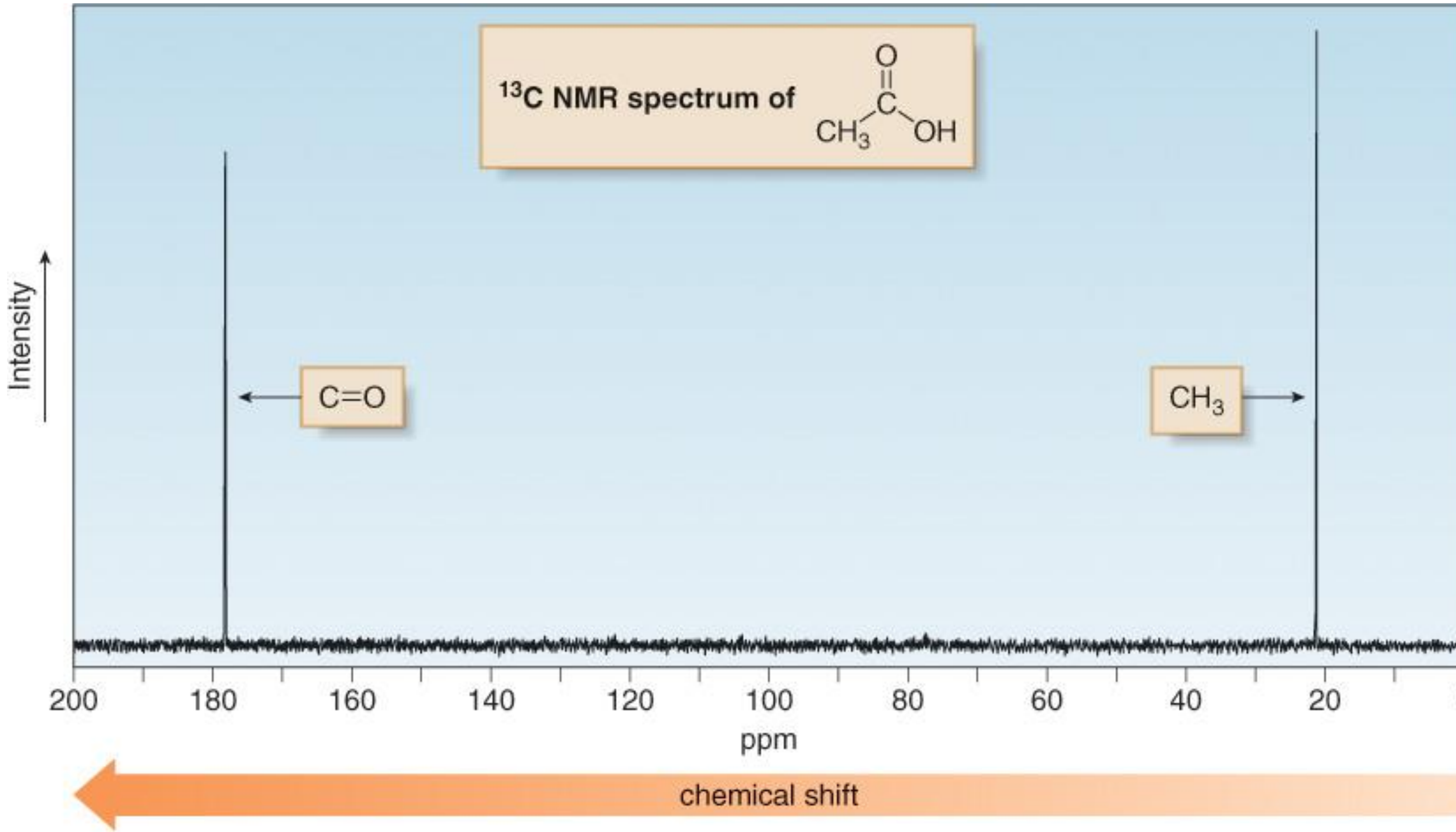
^{1}H NMR—Enantiotopic and Diastereotopic Protons.



- When substitution of two H atoms by Z forms enantiomers, the two H atoms are equivalent and give a single NMR signal. These two H atoms are called *enantiotopic* protons.

^{13}C NMR

^{13}C Spectra are easier to analyze than ^{1}H spectra because the signals are not split. Each type of carbon atom appears as a single peak.



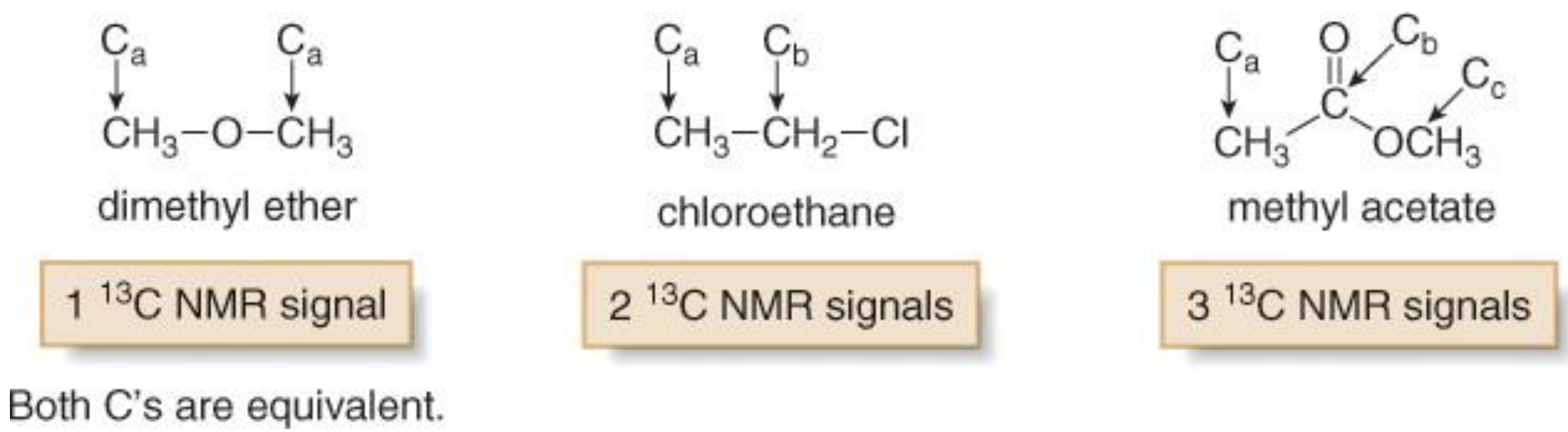
^{13}C NMR

- **The lack of splitting in a ^{13}C spectrum is a consequence of the low natural abundance of ^{13}C.**
- **Recall that splitting occurs when two NMR active nuclei—like two protons—are close to each other. Because of the low natural abundance of ^{13}C nuclei (1.1%), the chance of two ^{13}C nuclei being bonded to each other is very small (0.01%), and so no carbon-carbon splitting is observed.**
- **A ^{13}C NMR signal can also be split by nearby protons. This ^{1}H-^{13}C splitting is usually eliminated from the spectrum by using an instrumental technique that decouples the proton-carbon interactions, so that every peak in a ^{13}C NMR spectrum appears as a singlet.**
- **The two features of a ^{13}C NMR spectrum that provide the most structural information are the number of signals observed and the chemical shifts of those signals.**

Nuclear Magnetic Resonance Spectroscopy

^{13}C NMR—Number of Signals

- **The number of signals in a ^{13}C spectrum gives the number of different types of carbon atoms in a molecule.**
- **Because ^{13}C NMR signals are not split, the number of signals equals the number of lines in the ^{13}C spectrum.**
- **In contrast to the ^{1}H NMR situation, peak intensity is not proportional to the number of absorbing carbons, so ^{13}C NMR signals are not integrated.**

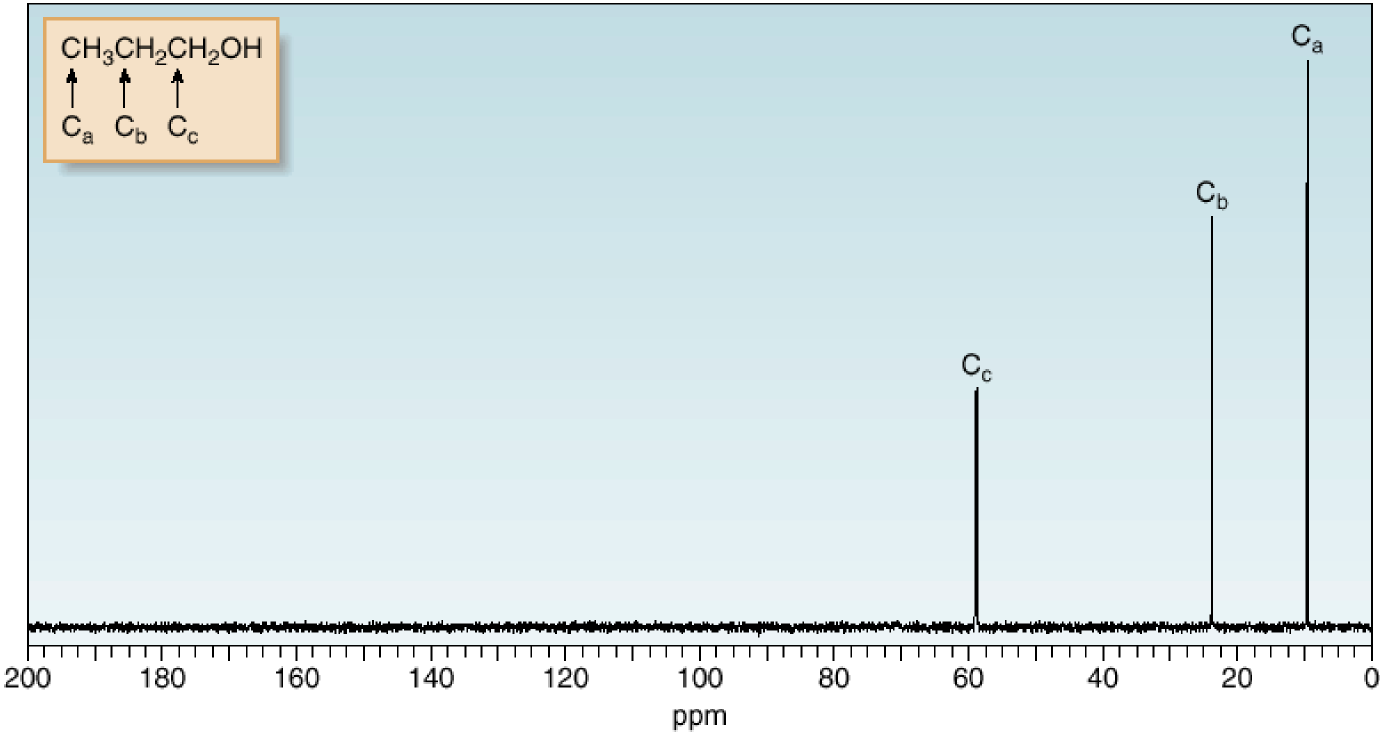


^{13}C NMR—Number of Signals

Figure 14.14
Representative ^{13}C NMR spectra

- The three types of C's in 1-propanol—identified as C_a, C_b, and C_c—give rise to three ^{13}C NMR signals.
- Deshielding increases with increasing proximity to the electronegative O atom, and the absorption shifts downfield; thus, in order of increasing chemical shift: $C_a < C_b < C_c$.

a. 1-Propanol



^{13}C NMR—Number of Signals

- The three types of C's in methyl acetate—identified as C_a, C_b, and C_c—give rise to three ^{13}C NMR signals.
- The carbonyl carbon (C_b) is highly deshielded, so it absorbs farthest downfield.
- C_a, an sp^3 hybridized C that is not bonded to an O atom, is the most shielded, and so it absorbs farthest upfield.
- Thus, in order of increasing chemical shift: $C_a < C_c < C_b$.

b. Methyl acetate